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The Origins and Diffusion of Watches in the Renaissance: Germany

At some unknown time and in some unknown place, an important development with long-lasting consequences took place: a miniaturised, portable timekeeper was developed. These small weight-driven clocks were already common by the middle of the fifteenth century, which is confirmed by an Italian illuminated manuscript from about 1460 that depicts the shop of a clockmaker with two weight-driven domestic clocks.⁴²⁹ For portable timepieces, the availability of an energy source independent of gravity was a necessary prerequisite. It seems likely that about 1400 the first spring-driven clocks were invented. Antonio Manetti (1423–1497), biographer of the famous architect Filippo Brunelleschi (1377–1446), stated that he made spring-driven clocks when working as a goldsmith. The passage has given rise to controversial discussions, but spring-driven clocks had probably been in use for many years by the time he was writing Brunelleschi's biography in about 1480.⁴³⁰

The verge escapement with foliot lacks isochronism, and therefore needed some sort of contrivance to equalize the varying traction force of the spring. To achieve this, the fusee⁴³¹, a cone-shaped pulley with a helical groove, was used, around which a gut string or chain attached to the mainspring barrel was wound. The idea

⁴²⁹ Modena, Biblioteca Estense: α.X.2.14 (Ms. Lat. 209), f. 11r; see: Francis Alan Burnett Ward, "A 15th century Italian Clockmakers' Workshop", *Antiquarian Horology and the Proceedings of the Antiquarian Horological Society* (12), 1980/81: 172–174. Clockmakers are depicted in this astronomical-astrological manuscript among "Mercury's children", i.e. people born in the hour ruled by the messenger of gods were said to become scientists, skilled artisans, artists and merchants, but also thieves and swindlers.

⁴³⁰ Giuseppe Brusa, "Early Mechanical Horology in Italy", *Antiquarian Horology and the Proceedings of the Antiquarian Horological Society* (18), 1989/90: 496–7.

⁴³¹ It seems that the word derives from the French *fusée* and late Latin *fusata* (spindle).

may have originated in military applications, since the *Bellifortis*, a manuscript of 1405 by Conrad Kyeser of Eichstätt shows a device for tensioning crossbows that employs it.⁴³² A miniature of c. 1450 to Heinrich Seuse's *Horologium sapientiae* depicts an octagonal movement with fusee⁴³³, and drawings by Leonardo da Vinci (1452–1519) with fusee constructions are extant.⁴³⁴ A drum-shaped clock made in 1525 by Jacob Zech has long been regarded as the earliest known application of the fusee⁴³⁵, but the earliest existing spring-driven clock with a fusee is the so-called *Burgunderuhr* (Burgundy clock) probably made for Philip the Good, Duke of Burgundy, in about 1435.⁴³⁶ The first archival evidence for the existence of spring-driven clocks is a manuscript written in Rome by Frater Paulus Almanus (Alemannus) c. 1475–85.⁴³⁷ It comprises thirty descriptions of clocks of a great variety, among them eight spring-driven timekeepers with **[p. 142]** fusees.⁴³⁸ Spring-driven clocks were made in increasingly smaller size. At the court of Ludovico Sforza, small, portable clocks (*horologiini piccoli et portativi*) and chiming timekeepers attached to ball costumes were mentioned in around 1490.⁴³⁹

⁴³² Göttingen, Niedersächsische Staats- u. Universitätsbibliothek: Cod. Ms. philos. 63, fol. 76v; see: Franz Maria Feldhaus, "Über den Ursprung von Federhaus und Schnecke", *Deutsche Uhrmacher-Zeitung* 54 (1930): 720–722.

⁴³³ Brussels, Bibliothèque Royale: Ms. IV, 111, fol. 13v.

⁴³⁴ Paris, Institut de France: Ms. B, fol. 50v (1488–90); Madrid, Biblioteca Nacional de España: Cod. Madrid I, fol. 85r.

⁴³⁵ E.g. Herbert Alan Lloyd, "The Origin of the Fusee", *The Antiquaries Journal* (31), 1951: 188–191; concerning the Zech clock (British Museum, on loan of the Society of Antiquaries) see W. H. Smyth, "Description of an Astrological Clock, belonging to the Society of Antiquaries of London: in a Letter to the President", *Archaeologia, or Miscellaneous Tracts Relating to Antiquity* (33), 1849: 8–35; Heinrich Otto, "The Famous Fusee Clock of Jakob Zech", *Horological Journal* (1930): 44–46, 62–64; (1931): 80–82, 117–120, 138–139, 158–160, 176–178, 220–223; Alfred Holniski, "Jacob Zech and a Royal Fusee Clock", *The Connoisseur* (152), 1963: 183–187.

⁴³⁶ Nuremberg, Germanisches Nationalmuseum: HG 9771; see Ernst v. Bassermann-Jordan, *Die Standuhr Philipps des Guten von Burgund* (Leipzig: Wilhelm Diebener, 1927; Max Engelmann, *Die Burgunder Federzuguhr um 1430* (Halle, Zentralverband der Deutschen Uhrmacher, 1927; Klaus Maurice, *Die deutsche Räderuhr: Zur Kunst und Technik des mechanischen Zeitmessers im deutschen Sprachraum* (Munich, C. H. Beck, 1976), vol. I, 85–87. This extraordinary timepiece belonged to the collection the Prince of Austria, Eduard von Collalto (1748–1833) and is mentioned first in 1835. Its authenticity has been disputed, but a set of technical drawings which originated in the environment of Jean Fusoris and Henri Arnaut de Zwolle in the fifteenth century show parallels in construction details. Such details would certainly be unknown to a forger of the nineteenth century.

⁴³⁷ State and Municipal Library of Augsburg: Cod. 2°, 209. The manuscript was first noticed by Ernst Zinner (see his publication: *Aus der Frühzeit der Räderuhr: Von der Gewichtsuhr zur Federzugsuhr* [Munich: R. Oldenbourg, 1954], 18–20) and published by Jan (John) Hendrik Leopold (*The Almanus Manuscript: Staats- und Stadtbibliothek Augsburg, Codex in 2° No. 209. Rome circa 1475–circa 1485* (London: Hutchinson, 1971).

⁴³⁸ Leopold, *The Almanus Manuscript*, 52–69, 90–99, 156–177, 194–205, 248–253.

⁴³⁹ Gerhard Dohrn-van Rossum, *History of the Hour: Clocks and Modern Temporal Orders*, tr. Thomas Dunlap (Chicago/London: The University of Chicago Press, 1996), 121.

In Germany, a few cities took the lead in establishing the manufacture of clocks and instruments. There often family dynasties were active, in which knowledge and skills were inherited from father to son. These included (inter alia) the Schissler and Klieber, later Miller and Tucher (all in Augsburg), Purmann and Schniep in Munich, Erasmus and Josua Habermel in Prague, Troschel and Reinmann in Nuremberg, as well as Christoph Trechsler the Elder and Younger at the court of the Elector of Saxony in Dresden.

In the course of the fifteenth century, Nuremberg became a centre of clock- and instrument making, the fine arts, as well as mathematical, astronomical and geographical studies. Located at the intersection of major trade routes, Nuremberg was (alongside Cologne) one of Germany's largest cities with extensive trade and a thriving hub of elaborate craftsmanship. Raw materials (copper, zinc, tin, brass and bronze) were readily available, and Nuremberg's metal craftsmen and instrument makers were renowned for the quality of their work, from which also mathematicians, astronomers and cartographers profited. In 1471, Johannes Regiomontanus (1436–1476) settled down in Nuremberg, and established an observatory and a printing shop, from which numerous mathematical and astronomical works emerged. His pupil and patron, Bernhard Walther (1530–1504), first employed a clock for astronomical observations.⁴⁴⁰

Augsburg also became a capital of the production of clocks and instruments.⁴⁴¹ As early as 1507 Ulrich Klieber, son of a coppersmith (*Kesselschmied*) and locksmith enjoyed a good reputation, which continued with his descendants to the beginning of the seventeenth century. Apart from multifunctional astronomical instruments (compendia, *Bestecke*) and sundials, the Klieber family built mechanical clocks.⁴⁴² Clocks and watches were also made by Johann Reinhold the Elder (c. 1550) and by the Buschmann dynasty in eight generations until the second half of the seventeenth century, starting with Caspar (I), who settled in Augsburg c. 1530.⁴⁴³

⁴⁴⁰ Donald deB. Beaver, "Bernard Walther: Innovator in Astronomical Observation", *Journal for the History of Astronomy* 1, 1970: 39–43.

⁴⁴¹ For the guild of clock- and watchmakers in Augsburg see Eva Groiss, „Das Augsburger Uhrmacherhandwerk“, in: Klaus Maurice and Otto Mayr (eds.), *Die Welt als Uhr: Deutsche Uhren und Automaten 1550–1650* (Munich/Berlin: Deutscher Kunstverlag, 1980), 62–89.

⁴⁴² Maximilian Bobinger, *Alt-Augsburger Kompaßmacher: Sonnen-, Mond- und Sternuhren, Astronomische und mathematische Geräte, Räderuhren* (Augsburg: Hans Rösler, 1966), 77–94.

⁴⁴³ Maximilian Bobinger, *Kunstuhrmacher in Alt-Augsburg* (Augsburg: Hans Rösler, 1969), 79–97; id., *Alt-Augsburger Kompaßmacher*, 133–147.

In the first half of the sixteenth century, clockmaking in Nuremberg was a free trade⁴⁴⁴, but some clockmakers joined the Locksmiths' Guild (*Plattschlosser*). In 1565 three clockmakers (Esaias Vogel, Hanns Praun and Markus Stopinger) were incorporated into the trade of the locksmiths, and requirements for their masterpieces were decreed by the City Council. The three petitioners, who were obviously experienced craftsmen, had to present a table clock striking the hours and the quarters, showing on one side the day and the night with sun and moon as well as quarters and twenty-four hours, on the other side a calendar, the planets (i.e., the days of the week) and the length of the day. The other specimen was a small watch striking the hours.⁴⁴⁵ But for all future applicants the table clock had to be furnished with an astrolabe in addition, and the watch should be equipped with an alarm and strike 12 or 24 hours. These timepieces had to be completed over a period of eight months.⁴⁴⁶

In the second half of the sixteenth century the manufacture of small, portable watches was obviously common in Nuremberg. Already in 1730 the astronomer

⁴⁴⁴ „Uff den bericht, das es des urmachens halben alhie kein gesetz noch ordnung hab, sonnder es ye unnd allwegen ein freye kunst gewest unnd noch sey, soll manns also, wie mann diesen bericht gefunden, statpflegern unnd den gehaimen räten zu Augsburg uff ir jüngst schreiben unnd begern also widerumb zuschreiben“ (Theodor Hampe, *Nürnberger Ratsverlässe über Kunst und Künstler im Zeitalter der Spätgotik und Renaissance [1449] 1474–1618 [1633]*, vol. I, [Vienna/Leipzig: K. Graeser & Kie., 1904], 498–9, nr. 3518).

⁴⁴⁵ „Zwen Uhren, das erst ein stehend Werk seiner Höhe am eisernen Geheuss sechs Zoll, an der Breite 4 ½ und der Dicke 2 ½ Zoll. Darin soll gemacht werden von Rad und Trieb von rechter Kunst ein Werk, das die 4 Viertel und die Stunden schlägt, auf der einen Seite soll es zeigen Tag und Nacht, als die Sonn- und Mondschein 24 Stund, und dass allweg dabei das Viertel und die Stund zu erkennen sei; auf der andern Seite soll sie haben den Kalender und die Planeten mit sammt der Tageslänge. Das andere Werk soll kleins Korn sein, wie man's pflegt am Hals zu tragen; darein soll gericht sein ein Werk, das zwölf schlägt und weckt“ (Jacob Stockbauer, *Nürnbergisches Handwerksrecht des XVI. Jahrhunderts: Schilderungen aus dem Nürnberger Gewerbeleben*, [Nuremberg: Verlag der Fr. Korn'schen Buchhandlung, 1879, 3; complete text of the order according to a copy of 1629 in Albert Gümbel, *Aktenstücke zur Geschichte der mittelfränkischen Uhrmacherkunst* (Halle: Zentralverband der Deutschen Uhrmacher, 1928), 1–8.

⁴⁴⁶ „2 Uhren, die erst ein stehendes Werk seiner Höhe, Breite und Dicke am Gehäuss wie das vorhingenannte, aber das Werk darin soll also verfertigt werden, dass es schlägt zwölf stund und dazwischen alle Viertel, und da man's haben will, so soll sie auch die 24 Stunden schlagen, die Minuten und dabei einen Wecker haben, sonderlich aber soll sie auf der einen Seiten die Tageslänge, den Kalender und die Planeten und auf der andern Seiten das Astrolabium mit seiner Zugehördt, dabei soll es zeigen die 12 Stunden und jedesmal die vier Viertel nebeneinander und soll solch Werk die Verborgenheit in sich haben, wenn man den Viertelzeiger reibt, dass die andern Zeiger alle gehen, und soll ein jeder das Messinggehäus samt dem Hut über der Glocken also machen, dass er dazu keine Patronen entlehnen soll, sondern die Zier, die einer in allem am Geheuss machen will, soll er pro forma selbst lassen schneiden, die hernach keiner dem andern leihe, sondern ein jeder soll solches von freier Hand selbst bewähren, ohne Jemandes Zuthun. Das andere Werk, das man am Halse trägt, soll dahin gerichtet sein, dass es schlägt 24 und 12 Stunden, wie man es haben will, darin soll auch ein Wecker sein, und solche 2 Stück soll einer bewähren in 8 Monaten“ (Stockbauer, *Nürnbergisches Handwerksrecht*, 3–4).

Johann Gabriel Doppelmayr (1677–1750) took it for granted that the invention of the pocket watch took place in Nuremberg, quoting the famous reference to Henlein in the *Cosmography* of Pomponius Mela edited by Johann Cochlaeus in 1512: *Each day more subtle things are invented. Thus Peter Hele, still quite a young man, makes works which even very learned mathematicians admire, for from a little bit of iron he makes clocks composed of many wheels which, however wound, will run without weights and show the time for 40 hours, and which may be carried on the person or in a purse.*⁴⁴⁷

But there are no objects extant which can be ascribed to Henlein with certainty.⁴⁴⁸ The history of the “invention” of the pocket watch is still a conglomerate of undisputed original sources and more or less audacious conjectures with a good dose of local patriotism.⁴⁴⁹

With its diversified and highly developed craftsmanship, Nuremberg seems to be a likely candidate for the creation of the first portable timekeeper. Moreover, the necessary tools and equipment required for the manufacture of small-sized mechanisms (small files, drills etc.) could be acquired easily.⁴⁵⁰ Little is still known about these “external” factors, however, and it is not unlikely that such favourable conditions may be encountered at other places too (e.g., Augsburg or Milan).

⁴⁴⁷ “*Inveniunt in dies subtiliora, etenim Petrus Hele juvenis adhuc admodum opera fecit, quae vel doctissimi admirantur Mathematici, nam ex ferro parva fabricat horologia plurimis digesta rotulis, quae quocunque vertantur, absque ullo pondere & monstrant et pulsant XL. horas, etiamsi in sinu marsupiove contineantur*” (*Historische Nachricht von den Nürnbergischen Mathematicis und Künstlern* [...] [Nuremberg: P. C. Monath, 1730], 286).

⁴⁴⁸ For a recent attempt, see: Dietrich Matthes, “A Watch by Peter Henlein in London?”, *Antiquarian Horology and the Proceedings of the Antiquarian Horological Society* (36), 2015: 183–194; by the same author: “Eine Uhr von Peter Henlein”, *Deutsche Gesellschaft für Chronometrie: Jahresschrift* (54), 2015: 35–44. The object is preserved in the British Museum (inv. nr. 1888, 1201.105), and Matthes’ observations appear to be definitive.

⁴⁴⁹ E.g. Enrico Morpurgo, *L’origine dell’orologio tascabile/The Origin of the Watch* (Rome: La Clessidra, 1954); id., *L’orologeria italiana dalle origini al Quattrocento* (Rome: La Clessidra, 1986); Jürgen Abeler, *In Sachen Peter Henlein* (Wuppertal: Selbstverlag, 1980). The late Giuseppe Brusa has aptly pointed out: “*The controversy concerning the invention of watches always owed more to national pride than to evidence*” (“Early Mechanical Horology in Italy”, 510; from the same author, see also: “I primi orologi da persona in Italia: Nuovi indizi e nuove eccellenti testimonianze”, *La Voce di Hora: Pubblicazione dell’Associazione Italiana Cultori di Orologeria Antica*, 1997, nr. 3, 3–20). For a sober, well-researched and balanced approach, see: Thomas Eser, *Die älteste Taschenuhr der Welt? Der Henlein-Uhrenstreit* (Nuremberg: Verlag des Germanischen Nationalmuseums, 2014); also id., “Die Henlein-Ausstellung im Germanischen Nationalmuseum: Rückblick, Ausblick, neue Funde”, *Deutsche Gesellschaft für Chronometrie: Jahresschrift* (54), 2015: 23–34.

⁴⁵⁰ Nuremberg has been designated as “*the ancestral city of filecutters*” (“*Ahnenstadt der Feilenhauer*”; see Otto Dick, *Die Feile und ihre Entwicklungsgeschichte* [Berlin: Julius Springer, 1925], 23–31).

Without doubt Henlein had a significant share in the development of portable timekeepers, but identifying him as the sole inventor and claiming Nuremberg as the one and only possible location is an inappropriate exaggeration. The miniaturisation of clock mechanisms seems to be a silent transition occurring at different places more or less contemporaneously.

Small and portable timepieces assumed different shapes in the sixteenth century. Cochlaeus does not specify the shape of Henlein's watches, but a record of payment of 1524 for a watch housed inside a musk apple (*Bisamapfel*) is extant and offers an important clue.⁴⁵¹ The "musk apple", better known as *pomander*, was a small pierced sphere in which one could keep amber or musk, which were believed to be preventive substances. During the Renaissance these objects [p. 143] were worn on one's clothes. It is thus likely that the first watches were of spherical shape, the earliest dated example being a pomander watch which formerly belonged to Philip Melanchthon (1497–1560). Where he acquired it is unknown and remains a point of conjecture.⁴⁵² Only two other authentic pomander watches⁴⁵³ and a fragment of a case ⁴⁵⁴ are extant. Watches, spherical in shape, were produced in various cities in South Germany in the first half of the sixteenth century, as well as in Blois.⁴⁵⁵ Drum-shaped clocks, of which about fifty examples are extant⁴⁵⁶, were made during the sixteenth century also, and Henlein may have produced small versions of these as well. The height of construction depended on the duration of run (i.e., the length/number of turns of the fusee). It is noteworthy that in a letter to Henlein

⁴⁵¹ „15 fl. H. Henlein für 1 verguldeten pyns Appfel für all Ding mit einem Oaiologium“; Albert Gümbel, *Peter Henlein der Erfinder der Taschenuhren* (Halle: Zentralverband der Deutschen Uhrmacher, 1924), 27, n. 29.

⁴⁵² Baltimore, The Walters Art Museum: 58.17; for a detailed survey see Maia Wellington Gahtan and George Thomas, „Philip Melanchthon's Watch Dated 1530“, *Antiquarian Horology and the Proceedings of the Antiquarian Horological Society* (26), 2001/02: 377–388 **[this note has been omitted in the printed version for unknown reasons]**.

⁴⁵³ Both in private collections, one of which is shown in this exhibition; see Ernst v. Bassermann-Jordan, „Ein zweiter Bisam-Apfel aus Peter Henleins Zeit“, *Die Uhrmacher-Woche* (31), 1924: 627–628). For a second pomander watch with unknown whereabouts see id., „Ein Bisam-Apfel aus Peter Henleins Zeit“, *Die Uhrmacher-Woche* (31), 1924: 301–302).

⁴⁵⁴ Alfred Beck, „Der Bisamapfel des Mathematisch-Physikalischen Salons“, *Die Uhrmacherwoche*, (49), 1942: 226–227 (badly damaged in World War II) **[n. 453 and 454 have been merged into one in the printed version]**.

⁴⁵⁵ A watch with spherical case in form of a terrestrial globe ascribed to Jacques de la Garde from Blois formerly belonged to the collection of the Time Museum in Rockford/Ill., which closed in 1999.

⁴⁵⁶ For a tabular compilation of extant examples see: Dietrich Matthes, „Corpus der tragbaren deutschen Dosenuhren des 16. Jahrhunderts“, in: Eser, *Die älteste Taschenuhr der Welt*, 205–219; id., „Die Gruppe der Henlein-Uhren“, in: *Anzeiger des Germanischen Nationalmuseums*, 2016 (forthcoming).

written c. 1530 by Kaspar von Schöneich, a councillor to the Duke of Mecklenburg, a chain is mentioned in connection with the fusee for the first time.⁴⁵⁷

In about 1550, the first neckwatches appeared. The application of the stackfreed, an alternative construction for the fusee, allowed somewhat flatter movements, the oldest dated example being a neckwatch of 1548 signed CW (Caspar Werner, Nuremberg?).⁴⁵⁸ But the construction itself is older; it has been applied in the alarum of a table clock of 1533⁴⁵⁹, and there are drawings of Leonardo extant.⁴⁶⁰

David Saul Landes (1924–2013) has remarked that the portable clock or watch proved to be a revolutionary instrument.⁴⁶¹ Only the most capable artisans could cope with the increasing requirements needed for the manufacture of minute components. Therefore, miniaturisation provided a technological stimulus, which was reinforced by the proliferation of timepieces and diffusion of ownership. Timekeepers, once the privilege and pleasure of an exalted few, were still very expensive and luxurious objects in the sixteenth century, but became available to a more and more widening circle of customers.

⁴⁵⁷ Peter Starsy, "Ein Brief von Kaspar Schöneich an Peter Henlein", *Uhren und Schmuck* (22), 1985: 23–27, 54–56 (especially 25–6; for two more letters sent from Henlein to Schöneich which have been found recently in Schwerin [Landeshauptarchiv] see: Thomas Eser, „Die Henlein-Ausstellung im Germanischen Nationalmuseum: Rückblick, Ausblick, neue Funde“, *Deutsche Gesellschaft für Chronometrie: Jahresschrift* [54], 2015: 31–33. Gruet of Geneva is commonly credited with the introduction of the fusee chain (e.g. Willis Isbister Milham, *Time & Timekeepers Including the History, Construction, Care and Accuracy of Clocks and Watches* [New York: Macmillan, 1941], 126, 224, 541).

⁴⁵⁸ Formerly in possession of Jürgen Abeler (Uhrenmuseum Wuppertal), now in the Patek Philippe Museum in Geneva.

⁴⁵⁹ Copenhagen, Nationalmuseet: inv. nr. 2164.

⁴⁶⁰ Leonardo's drawings show a horizontal construction as well as a vertical one (Dietrich Matthes, „A Watch by Peter Henlein in London?“, *Antiquarian Horology and the Proceedings of the Antiquarian Horological Society* [36], 2015: 185; Giuseppe Brusa and John [Jan] Hendrik Leopold, "The Vertical Stackfreed: Mechanical Evidence and Early Eye-Witnesses", *Nuncius: Annali della storia della scienza* [12], 1997: 420–422). For a collection of 23 watches and clocks with stackfreed from the sixteenth and seventeenth century exhibited in the Musée de l'horlogerie et de l'émaillerie de Genève, see: Eugen Gschwind, *Stackfreed 1540–1640* (Basle: s. n., 1979).

⁴⁶¹ David Saul Landes, *Revolution in Time: Clocks and the Making of the Modern World* (Cambridge, Mass./London: The Belknap Press of Harvard University Press, 1983), 87–88.